

Tarea sección 10.2

$$9) \frac{(x-3)^2}{16} + \frac{(y+4)^2}{9} = 1$$

$$h = 3, K = -4$$

$$C = (3, -4)$$

$$\frac{(x-h)^2}{a^2} + \frac{(y-K)^2}{b^2} = 1$$

$$a^2 = 16 \Rightarrow a = 4$$

$$b^2 = 9 \Rightarrow b = 3$$

$$V_1 = (h-a, K) = (-1, -4)$$

$$V_2 = (h+a, K) = (7, -4)$$

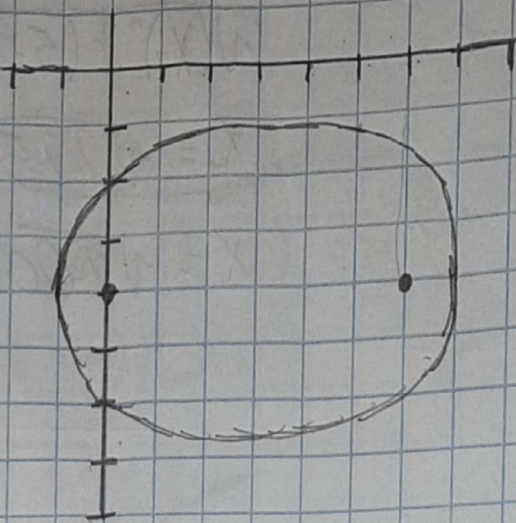
$$C^2 = a^2 - b^2$$

$$C = \sqrt{16 - 9}$$

$$C = \sqrt{7}$$

$$F_1 = (h-C, K) = (3-\sqrt{7}, -4)$$

$$F_2 = (h+C, K) = (3+\sqrt{7}, -4)$$



$$13) 25x^2 + 4y^2 - 250x - 16y + 541 = 0$$

$$16\left(\frac{1}{2}\right)^2 = 25$$

$$25x^2 - 250x + 4y^2 - 16y = -541$$

$$25(x^2 - 10x + 25) + 4y^2 - 16y = -541 + 25(25)$$

$$25(x-5)^2 + 4y^2 - 16y = 84$$

$$25(x-5)^2 + 4(y^2 - 4y) = 84$$

$$25(x-5)^2 + 4(y^2 - 4y + 4) = 84 + 4(4)$$

$$25(x-5)^2 + 4(y-2)^2 = 700$$

$$\frac{(x-5)^2}{4} + \frac{(y-2)^2}{25} = 7$$

$$C = (5, 2)$$

$$a = 2$$

$$b = 5$$

$$c = \sqrt{2^2 - 5^2}$$

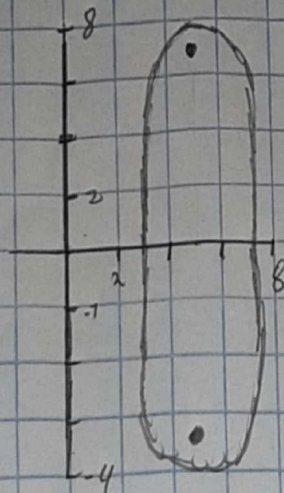
$$c = \sqrt{21}$$

$$V_1 = (h, k-b) = (5, -3)$$

$$V_2 = (h, k+b) = (5, 7)$$

$$F_1 = (h, k-c) = (5, 2 - \sqrt{21})$$

$$F_2 = (h, k+c) = (5, 2 + \sqrt{21})$$



$$17) V_1(-7, 1), V_2(3, 1)$$

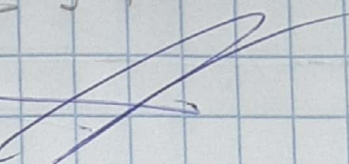
$$M_1(-2, -1), M_2(-2, 3)$$

$$C(-2, 1)$$

$$y = 2^2 = 4$$

$$x = 5^2 = 25$$

$$\frac{(x+2)^2}{25} + \frac{(y-1)^2}{4} = 1$$



$$37) e = \frac{1}{2} + c$$

$$P(7,3)$$

V = on eje X

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{1}{a^2} + \frac{3}{b^2} = 1$$

$$\frac{1}{2} = \frac{\sqrt{b^2 - a^2}}{b}$$

$$\frac{b}{2} = \sqrt{b^2 - a^2}$$

$$\frac{x^2}{13} + \frac{4y^2}{39} = 1$$

$$\frac{b^2}{4} = b^2 - a^2$$

$$a^2 = b^2 - \frac{b^2}{4} =$$

35) Eje > horizontal longitud 8, eje menor 5

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$8/2 = 4^2 = 16$$

$$5/2 = \frac{5^2}{2} = \frac{25}{2}$$

$$\frac{x^2}{16} + \frac{4y^2}{25} = 1$$

$$43) F(0, 15), F'(0, -15)$$

$$K = 39$$

$$\frac{x^2}{64} + \frac{y^2}{289} = 1$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$C = \sqrt{64^2 - 289^2}$$

$$C = \sqrt{281.82}$$

$$F = (64, 28.82)$$

$$C = (64, 28.82)$$

47) Mital izquierdo $\frac{x^2}{36} + \frac{y^2}{25} = 1$

$$x = \sqrt{36 \left(\frac{y^2}{25} \right) + 7}$$

$$x = \frac{6}{5} \sqrt{25 - y^2}$$

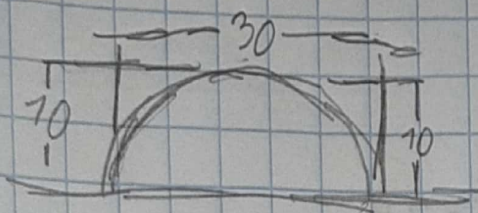
$$x = -\frac{6}{5} \sqrt{25 - y^2}$$

57) $y = 2 - 7 \sqrt{1 - \frac{(x+1)^2}{9}}$

$$\frac{(x+1)^2}{9} = -\frac{(y-2)^2}{9} + 1$$

$$\frac{(x+1)^2}{9} + \frac{(y-2)^2}{9} = 1 \text{ mited inferior}$$

59)



$$\frac{x^2}{75^2} + \frac{y^2}{10^2} = 1$$

$$x = \frac{36}{225} + \frac{y^2}{100} = 1$$

$$y^2 = 100 \left(1 - \frac{36}{225} \right) = 84$$

$$y = \sqrt{84} \text{ pies}$$